

Depu Meng, Ph. D.

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Research Engineer at Applied Intuition

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Research Interests

World foundation model: Robotics world model, video generation, world action model, multi-modal foundation model pretraining, end-to-end autonomous driving.

Education

University of Science and Technology of China - Microsoft Research Asia	BEIJING, CHINA
Ph. D. in Control Science and Engineering	Sept. '18 – Jun. '23
• Advisors: Dr. Baining Guo (Microsoft), Prof. Houqiang Li (USTC)	
University of Science and Technology of China	HEFEI, CHINA
B.E. in Electrical Engineering (School of Gifted Young)	Sept. '14 – Jun. '18

Work Experience

Applied Intuition	MOUNTAIN VIEW, CA
Research Engineer, Research	Sep. '25 – present
• Physical world model, video generation, 3D foundation model, scene reconstruction, and end-to-end driving foundation models.	
DiDi Autonomous Driving	SAN JOSE, CA
Machine Learning Engineer, AI Research	Dec. '24 – Aug. '25
• End-to-end driving model training and 3d reconstruction with Gaussian Splatting.	
University of Michigan	ANN ARBOR, MI
Research Fellow, Michigan Traffic Lab	Aug. '23 – Dec. '24
• Near-miss detection and road-side perception.	
• Led the team won USDOT Intersection Safety Challenge Stage 1A and 1B.	

Internship Experience

University of Michigan	ANN ARBOR, MI
Research Assistant, Michigan Traffic Lab	Apr. '22 – Aug. '23
• Image generation and driving scenario simulation.	
Mentor: Prof. Henry X. Liu	
Meituan	BEIJING, CHINA
Intern, Autonomous Delivery Group	Aug. '21 – Apr. '22
• Joint Lidar-based object detection and motion prediction.	
Mentor: Dr. Changqian Yu	
Microsoft Research Asia	BEIJING, CHINA
Intern, Visual Computing Group	Jul. '19 – Jul. '21
• Transformer-based object detection and unsupervised pretraining.	
Mentor: Dr. Jingdong Wang	
Microsoft Research Asia	BEIJING, CHINA
Intern, Visual Computing Group	Jul. '17 – Jul. '18
• CNN architecture design and image classification.	
Mentor: Dr. Jingdong Wang	

Publications

* *equal contribution*, † *project lead*.

LaMo: Self-Supervised Latent Motion Priors for Physical Realism in Video Generation

Bo Jiang, **Depu Meng**, Yihan Hu, Yichen Xie, Tianshuo Xu, Wei Zhan

Submitted

Teaching Video Generators to Remember: Eliciting Dynamic Memory for Out-of-Sight State Evolution

Tianshuo Xu, Yichen Xie, **Depu Meng**†, Chensheng Peng, Quentin Herau, Bo Jiang, Yihan Hu, Wei Zhan

Submitted

URoPE: Universal Relative Position Embedding across Geometric Spaces

Yichen Xie, **Depu Meng**, Chensheng Peng, Yihan Hu, Quentin Herau, Masayoshi Tomizuka, Wei Zhan

Submitted

SpectralSplat: Appearance-Disentangled Feed-Forward Gaussian Splatting for Driving Scenes

Quentin Herau*, Tianshuo Xu*, **Depu Meng**†, Zhijie Yang, Chensheng Peng, Spencer Sherk, Yihan Hu, Wei Zhan

Submitted

Exploring Communication and Roadside Perception Requirements for Cooperative Warning Systems at Intersections

Tinghan Wang, **Depu Meng**, Boqi Li, Rusheng Zhang, Yukun Zuo, Shengyin Shen, Darian Hogue, Michael Maile, Michael Shulman, Henry X. Liu

IEEE Intelligent Vehicles Symposium, 2025

Towards Comprehensive Roadside Intelligence: Sensor Fusion and Full-stack Perception with Multiple Cameras

Rusheng Zhang, **Depu Meng**, Shengyin Shen, Boqi Li, Tinghan Wang, Henry X. Liu

IEEE Intelligent Vehicles Symposium, 2025

MSight: An Edge-cloud Infrastructure-based Perception System for Connected Automated Vehicles

Rusheng Zhang*, **Depu Meng***, Shengyin Shen, Zhengxia Zou, Houqiang Li, Henry X. Liu

Tech Report.

Robust Roadside Perception for Autonomous Driving: An Annotation-free Strategy with Synthesized Data

Rusheng Zhang, **Depu Meng**, Lance Bassett, Shengyin Shen, Zhengxia Zou, Henry X. Liu

IEEE Transactions on Intelligent Vehicles, 2024

Systematic Assessment of Roadside Perception Systems for Automated Vehicles: Insights from Field Testing

Rusheng Zhang, **Depu Meng**, Tinghan Wang, Tai Karir, Shengyin Shen, M. Maile, M. Shulman, Henry X. Liu

Transportation Research Board Annual Meeting, 2024

ROCO: A Roundabout Traffic Conflict Dataset

Depu Meng, Owen Sayer, Rusheng Zhang, Shengyin Shen, Houqiang Li, Henry X. Liu

Transportation Research Board Annual Meeting, 2023

HyMo: Hybrid Motion Representation Learning for Prediction from Raw Sensor Data

Depu Meng, Changqian Yu, Deheng Qian, Houqiang Li, Dongchun Ren

IEEE Transaction on Multimedia, 2023

CORE: Consistent Representation Learning for Face Forgery Detection

Yunsheng Ni, **Depu Meng**, Changqian Yu, Chengbin Quan, Dongchun Ren, Youjian Zhao

CVPR 2022 Workshop on Media Forensics

Conditional DETR for Fast Training Convergence

Depu Meng*, Xiaokang Chen*, Zejia Fan, Yuhui Yuan, Gang Zeng, Houqiang Li, Lei Sun, Jingdong Wang
International Conference on Computer Vision, 2021

Consistent Instance Classification for Unsupervised Representation Learning

Depu Meng, Zigang Geng, Zhirong Wu, Bin Xiao, Houqiang Li, Jingdong Wang
ICCV 2021 Workshop on Self-supervised Learning for Next-Generation Industry-level Autonomous Driving

Bottom-Up Human Pose Estimation by Ranking Heatmap-Guided Adaptive Keypoint Estimates

Ke Sun, Zigang Geng, **Depu Meng**, Bin Xiao, Dong Liu, Zhaoxiang Zhang, Jingdong Wang
Tech Report

Deep Convolutional Neural Networks with Merge-and-Run Mappings

Liming Zhao, Mingjie Li, **Depu Meng**, Xi Li, Zhuowen Tu, Zhaoxiang Zhang, Yueting Zhuang, J. Wang
International Joint Conference on Artificial Intelligence, 2018

Awards

U.S. DOT Intersection Safety Challenge Stage 1B Tier 1 Winner Team (Prize: \$750,000)	Jan. '25
U.S. DOT Intersection Safety Challenge Stage 1A Winner Team (Prize: \$100,000)	Jan. '24
Shenzhen Stock Exchange Scholarship, USTC	Dec. '22
Star of Tomorrow Internship Award, Microsoft Research Asia	Jul. '18
First Prize in Intelligent Robot Competition, Harbin Institute of Technology	Jul. '16
The AEGON-INDUSTRIAL Fund Scholarship, USTC	Oct. '15

Services

Conference Reviewer: CVPR, ECCV, ICCV, NeurIPS, ACM MM, TRBAM, IROS 2023

Journal Reviewer: IJCV, IEEE T-IV, IEEE T-MM, IEEE T-CSVT, Neurocomputing, Pattern Recognition